NICHOLAS ROSS

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EDUCATION

University of Notre Dame, Notre Dame, IN — Expected May 2027
Ph.D. in Bioengineering, Biophotonics Lab (PI: Dr. Thomas O'Sullivan)
GPA: 3.77/4.0

University of South Florida, Tampa, FL — Aug 2012
B.A. in Communications

RESEARCH EXPERIENCE

University of Notre Dame, Notre Dame, IN — Aug 2021 – Present
Graduate Research Assistant, Biophotonics Lab

Project: Detecting Traumatic Brain Injury using NIRS — Jul 2025 – Present

- Designed a frequency-domain near-infrared spectroscopy (FD-NIRS) probe and tissue mimicking phantoms (physical and simulated) for noninvasive detection of traumatic brain injury, specifically hematomas.
- Developed Monte Carlo, finite element method (FEM), and diffusion models of multilayer tissue in Python to quantify where diffusion theory fails for highly absorbing hematomas, to enable estimation of hematoma oxy- and deoxyhemoglobin with FD-NIRS.
- Accelerated optical property recovery in MATLAB by replacing an exhaustive residual-minimization search over a 6.25M-entry look-up table with k-nearest-neighbor search, achieving a 10x speedup, and later introduced a trust-region algorithm to reduce computational load on the front end.
- Wrote ESP32 firmware (C++) to synchronize RF/DC switching, VNA modulation sequencing, and laser diode current control while acquiring and streaming measurement data.
- Evaluated PCB RF performance from 50–400 MHz for insertion loss, SNR, and noise using VNA and oscilloscope.
- Model, create, and characterize silicone-based tissue-mimicking phantoms to simulate hematomas at varying depths in cerebral tissue and evaluate hematoma detection capability of NIRS device.

Project: Wearable Frequency Domain-NIRS Device — Sept 2024 – Jan 2026

- Built MATLAB programs for real-time optical property recovery and visualization of biological signals.
- Validated custom-built NIRS devices through *in-vivo* cognitive (e.g., mental arithmetic) and physiological (e.g., Valsalva maneuver) evaluations against established results.
- Implemented signal processing methods, including filtering, Fourier Transform, and power spectral density analysis, to enhance data quality and extract physiologically relevant information (e.g., heart rate and respiration).
- Utilized profiling tools to identify and resolve performance bottlenecks in hardware interfacing scripts, increasing system sampling rate from 5 to 20 Hz.

Project: Novel Phased-Array Method for Deeper Brain Imaging — Jun 2023 – Sept 2024

- Constructed light transport simulations using FEM and Monte Carlo models to evaluate depth sensitivity and tomography capabilities of novel method for functional NIRS (fNIRS).

Feinstein Institute for Medical Research, Manhasset, NY — Jun – July 2024
Graduate Research Assistant, Translational Neurophysiology Laboratory
Principal Investigator: Dr. Stavros Zanos

- Analyzed Utah-array recordings (Intan) in anesthetized swine during vagus nerve stimulation, using Wave_clus spike sorting to relate neuronal firing patterns to evoked cardiac and respiratory responses.
- Assisted in implantation of Utah-Array and nerve-cuff placement surgeries in swine and murine preclinical studies.
- Supported murine preclinical studies of ultrasound-based peripheral nerve neuromodulation for inflammation.

University of Central Florida, Orlando, FL — Oct 2019 – Feb 2021
Undergraduate Research Assistant, Bioelectronics Lab
Principal Investigator: Dr. Brian Kim

- Created intracranial implant to house a 10x10 tetrode array for recording neural activity.

SKILLS

Coding: Arduino, bash, C/C++, MATLAB, Python

Software: Autodesk Inventor and SolidWorks (3D CAD), Altium and KiCAD (PCB design), Claude Code (agentic), MCX (Monte Carlo), NIRFAST (FEM), TracePro (ray tracing), TRVNA (vector network analyzer), Wave_clus (neuron spike sorting)

Hardware: Arduino, Avalanche Photodiodes, ESP32, FDM and Resin/Polyjet 3D Printers, Laser Driver, Oscilloscope, Silicon Photomultipliers, Spectrum Analyzer, Vector Network Analyzer

PUBLICATIONS & PRESENTATIONS

1. Lahade S, **Ross NA**, and O'Sullivan TD, "Advances in fully integrated wearable near-infrared spectroscopy: hardware innovations and biomedical applications", J. Phys. Photonics 8 022004. (Co–First Author, Paper)
2. Lahade S, Yazdi S, **Ross NA**, Smith D, Mohammadi S, Green MM, and O'Sullivan TD, "Wearable frequency–domain near–infrared spectroscopy (FD–NIRS) system", Biomed. Opt. Express 17, 555-571 (January 2026). (Paper)
3. Dale R, **Ross NA**, Howard S, O'Sullivan TD, and Dehghani H, "Towards real–time diffuse optical tomography with a handheld scanning probe", Biomed. Opt. Express 16, 1582-1601 (March 2025). (Paper)
4. Stute K, Gosse L, Montero–Hernandez S, Perkins G, Yucel M, **Ross NA** et. al. "The fNIRS glossary project: a consensus–based resource for functional near–infrared spectroscopy terminology", Neurophotonics (March 2025). (Paper)
5. **Ross NA**, Abdalsalam O, Howard S, Dehghani H, and O'Sullivan TD, "Towards a wearable, fibre–free frequency–domain near–infrared spectroscopy device using structured interrogation", functional Near–Infrared Spectroscopy (fNIRS) 2024 International Conference (September 2024). (Presentation)
6. Dale R, Zheng B, Orihuela–Espina F, **Ross NA**, O'Sullivan TD, Howard S, and Dehghani H, "Deep learning–enabled high–speed, multi–parameter diffuse optical tomography", J. Biomed. Opt. 29(7) 076004 (July 2024). (Paper)
7. Stillwell R, Longoria A, Gonzalez KA, Sheeder EW, **Ross NA**, and O'Sullivan TD. "Demonstration of an entirely handheld real–time frequency–domain near–infrared spectroscopy imaging system", Proc. SPIE PC12827, Multiscale Imaging and Spectroscopy V, PC128270A (March 2024). (Presentation)
8. Abdalsalam O, **Ross NA**, Perkins GA, Dale R, Dehghani H, O'Sullivan TD, Ferradal SL, "Comparison of multi-distance and multi-frequency methods in frequency-domain near-infrared spectroscopy", Proc. SPIE 12376, Optical Tomography and Spectroscopy of Tissue XV, 1237607 (March 2023)

AWARDS, GRANTS, & FELLOWSHIPS

BR41N.IO BCI Designer's Hackathon – BCI Data Analysis 3rd Overall Apr 2026
g.tec medical engineering GmbH

James L. Massey Travel Grant Jan 2025
University of Notre Dame, Electrical Engineering Dept.

Graduate School Professional Development Award Sept 2024
University of Notre Dame Graduate School

Precision Medicine Research Fellowship Jun – Jul 2024
Berthiaume Institute for Precision Health

McCloskey New Venture Business Competition May 2024
Visilocate – Breast tumor localization device
Grand prize winner and 6 additional awards ($75,000 total)

Innovative Programs to Enhance Research Training (IPERT) Jul 2023
National Institute of General Medical Sciences

Rapid Exposure to Advanced Computational Training (REACT) Jun 2023
Harper Cancer Research Institute

Graduate Research Fellowship Program (GRFP) - Not awarded Mar 2023

PROFESSIONAL TRAINING

Multiscale Biophotonics: A Platform for Interdisciplinary Research Training and Career Development Jul 9 – 20, 2023
University of California–Irvine, Irvine, CA

Harper Cancer Center Strategic Management Initiative Jun 5 – Jul 7, 2023
University of Notre Dame, Notre Dame, IN

OTHER PROFESSIONAL EXPERIENCE

Vizergy Digital Marketing, Jacksonville, FL Feb 2018 – Mar 2020
Account Manager & Search Engine Optimization Specialist

- Handled marketing and SEO strategy for 20+ Hyatt hotels.

Aurotech Inc., Silver Springs, MD (Remote) Jun 2015 – Feb 2018
Outside Sales & Marketing Manager

- Managed small sales team and coordinated all outbound sales and marketing efforts.

Aerotek, Jacksonville, FL Oct 2012 – Jun 2015
Senior Recruiter

- Recruited and placed candidates across healthcare and financial services roles by sourcing talent, screening for technical and cultural fit, and guiding hires from first outreach through offer acceptance.